

Assessment of Cycloxaprid Toxicity: Oxidative Stress and Cell Death Mechanisms in 4T1 Epithelial Mouse Mammary Cells

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Abstract

Global warming has resulted in a rise of pest populations and pesticide resistance, driving the widespread use of pesticides. This has significantly increased food and public safety concerns. Among the most prevalent pesticides are neonicotinoids, known for their environmental persistence and bioaccumulation, which ultimately pose potential health risks to humans. Cycloxaprid is a novel neonicotinoid insecticide which is predicted to dominate future agricultural practices as it has a unique chemical structure that heightens its efficacy against resistant pests. However, cycloxaprid has not been thoroughly assessed for carcinogenic properties. This study examined live and dead cell counts at various levels of cycloxaprid exposure, as well as a DNA laddering assay to observe DNA damage. Through these experiments, the research aimed to determine whether cycloxaprid induces cancer through DNA damage via oxidative stress, specifically in 4T1 epithelial mouse mammary cells.

Contrary to this initial hypothesis that predicted uncontrolled cell growth indicative of cancer, cycloxaprid exposure caused a significant decline in cell viability and promoted cell death. Furthermore, cycloxaprid exposure resulted in DNA fragmentation and smearing, which are signs of apoptosis and necrosis. These results were likely caused by excessive reactive oxygen species that were produced when cycloxaprid interacted with mitochondrial nicotinic acetylcholine receptor. The findings of this study suggest that cycloxaprid is cytotoxic rather than carcinogenic.

Introduction

The escalating impacts of climate change has led to a surge in food safety concerns; global warming has caused a rise in pest populations and pesticide resistance, leading to excessive and improper use of pesticides (Duchenne-Moutien & Neetoo, 2021). Among these, neonicotinoids-the most widely used class of insecticides-are known to persist in the environment and bioaccumulate in living organisms, ultimately entering the food chain and posing significant health risks for humans (K. Chen et al., 2021; Kaur et al., 2024; Z. Li, 2020; Ray & Shaju, 2023). These insecticides, including cycloxaprid, a novel neonicotinoid, have been linked to various toxic effects, such as reproductive and developmental harm and increased cancer risk (Cui et al., 2020; Y. Zhang et al., 2023).

As cancer is one of the leading causes of death, responsible for millions of new diagnoses every year, identifying environmental and chemical contributors to its development is crucial for preventing future cases and improving public health outcomes (Siegel et al., 2024). Understanding the causes of cancer, including potential links to widespread pesticide exposure, could provide vital insights into reducing the burden of this disease. Of particular concern is cycloxadprid, a novel neonicotinoid insecticide. Neonicotinoids are the most popular type of insecticides, taking up around 27% of the global insecticide market (Stehle et al., 2023). Out of this, cycloxadprid, being used in agriculture as a replacement for traditional neonicotinoids, is predicted to become the next most commonly used neonicotinoid (Cui et al., 2020; Dong et al., 2022; Luo et al., 2024). Cycloxadprid is known to accumulate in plant tissues and have been detected in the bodies of rural populations, especially children (Tao et al., 2019; Tsegay et al., 2024). Furthermore, researchers have found neonicotinoid insecticides in 53% of surface water collected from streams; given this large presence of cycloxadprid in plants and various environments, understanding cycloxadprid's potential carcinogenic effects, especially through mechanisms like oxidative stress, is crucial. (X. Cheng et al., 2019; Ray & Shaju, 2023). This research aims to understand the possible carcinogenicity of cycloxadprid, to support efforts to mitigate the associated public health risks.

Bioaccumulation of Toxic Chemicals in Plants and Organisms

Bioaccumulation refers to the gradual accumulation of toxins, such as heavy metals or pesticides in living organisms (Gómez-Regalado et al., 2023; Malmberg et al., 2021). It is essentially when the rate of absorption of certain chemicals is higher than the rate of excretion, meaning the chemical has built up in the organism (Wieczorek et al., 2023). Many chemicals are additionally persistent in the environment, consequently entering organisms at the top of the food chain; because many of these chemicals are lipophilic*, they are able to remain in lipids of living organisms (Malmberg et al., 2021). For example, mercury builds up in large fish such as tuna, and DDT, once a widely utilized pesticide, still persists in the aquatic food web decades after its ban (Barbosa et al., 2022; Sun et al., 2022; Turusov et al., 2002). The accumulation of these toxins is detrimental to not only such organisms but also to humans, even causing cancer. One example is heavy metals absorbed in fish, which are transferred to humans when eaten, and the toxic metal bioaccumulates in colon cells to cause colorectal cancer (Bonfiglio et al., 2024; Okereafor et al., 2020).

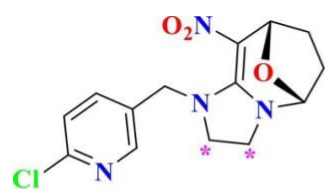
Toxins enter plants through two main locations, the soil or the air. Many chemicals accumulate in surface layers of soils either from atmospheric air or contaminated water (Azeh Engwa et al., 2019; Geng et al., 2019; Wieczorek et al., 2023). When the roots of the plants conduct passive transport to intake nutrients from the soil, contaminants in the dirt are also absorbed (Azeh Engwa et al., 2019; Doucette et al., 2018). On the other hand, the stomata and cuticles in the leaves are pathways that the plant can absorb chemicals from the air (Collin et al., 2022; Doucette et al., 2018). Especially with pesticides, which are often sprayed onto plant leaves, the exchange with air is a large source of pesticide exposure into the plant (Li, 2020). Once inside the plant, the movement and distribution of such contaminants are dependent on the properties of the chemical and the plant (Doucette et al., 2018). While some chemicals may be filtered out by metabolization or evaporating from plant surfaces, contaminants can also move through the xylem or phloem to different parts of the plant (Azeh Engwa et al., 2019; Collin et al., 2022; Doucette et al., 2018). Contaminants may migrate to edible parts of the plant; pesticide residues can then bioaccumulate in the fatty tissues of animals that consume plants, transporting the chemicals up the food chain, with the potential to enter human bodies (Li, 2020).

Vocabulary Key

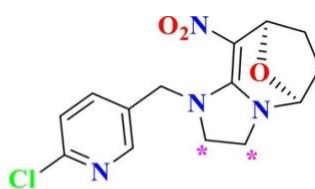
Lipophilic* - Able to be dissolved in lipids (fats)
Cis configuration* - Two similar groups on a molecule are on the same side of a central plane (Kauffman & Halpern, n.d.)
Trans configuration* - Two similar groups on a molecule are on opposite sides of a central plane (Kauffman & Halpern, n.d.)
Foliar absorption* - The process where plants intake water/nutrients through their leaves (Fernández et al., 2021)
Enantiomers* - A pair of molecules that are mirror images of one another but are non-superimposable (Hao, 2024)
Hepatotoxicity* - Damage to the liver caused by chemicals
Genotoxicity* - Damage to DNA caused by chemicals
Nicotinic acetylcholine receptors (nAChRs)* - Type of receptor in the nervous system that plays a key role in fast synaptic communication in response to acetylcholine (Barrantes, 2023)
Reactive oxygen species (ROS)* - Unstable molecules containing oxygen. At excessive levels, they have harmful effects (J. Liu et al., 2023)

Chemical Properties of Neonicotinoids and Cycloxaprid

Neonicotinoids are the most widely used class of insecticides in the world, with the first neonicotinoid being imidacloprid, introduced in the 1990s (Cui et al., 2020; Siegel Mph et al., 2024; Y. Zhang et al., 2023). However, as neonicotinoids gained widespread use, pests gradually developed resistance, necessitating alternative solutions (X. Cheng, Zhang, Wang, et al., 2022; Cui et al., 2020). In response, cycloxaprid was created, featuring a unique cis configuration* that differs from the trans configuration* found in all other commercialized



¹⁴C-5R,8S-CYC



¹⁴C-5S,8R-CYC

Figure 1. Chiral chemical structures of cycloxaprid enantiomers, and the positions of ¹⁴C labeling are marked with asterisks (X. Cheng, Zhang, Shao, et al., 2022)

neonicotinoids (Cui et al., 2018a, 2020). Cycloxaprid is a chiral neonicotinoid with two enantiomers, 5S8R-CYC and 5R8S-CYC, which both enter and act in plants similarly (X. Cheng, Zhang, Shao, et al., 2022). As shown in Figure 2, cycloxaprid is an oxabridged cis-

nitromethylene neonicotinoid, meaning its chemical structure contains a nitro (NO₂) group in a cis-configuration (X. Cheng, Zhang, Shao, et al., 2022; Y. Zhang et al., 2023). This cis configuration has a unique spatial arrangement, which enhances its binding affinity to insect receptors and makes it effective against pests resistant to conventional neonicotinoids which have a trans configuration (Cui et al., 2020; Dong et al., 2022; Y. Zhang et al., 2023). Neonicotinoids, in general, contain a nitro (NO₂) or cyano (CN) group, which are positively charged and thus exhibit binding affinity to the nicotinic acetylcholine receptors in insects, which are negatively charged (Alhalhooly & Sine, 2024; Niessen, 2021; Schindler, 2021; Q. Wang et al., 2023).

Like these conventional neonicotinoids, CYCs targets the nicotinic acetylcholine receptors in the nervous system of insects but binds to them differently (X. Cheng et al., 2019; Y. Zhang et al., 2023). As a result, cyclozaprid is one of the most effective insecticides used in agriculture (Cui et al., 2020; Liu et al., 2015; Y. Zhang et al., 2023).

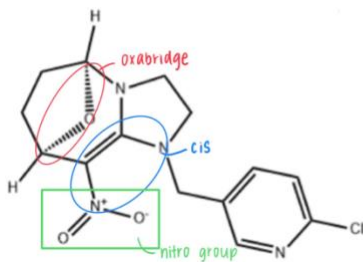


Figure 2a. Cyclozaprid, oxabridged cis-nitromethylene neonicotinoid (Q. Wang et al., 2023)

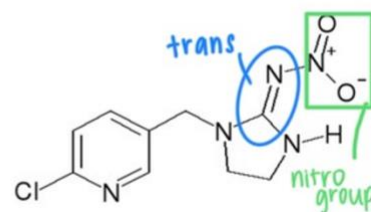


Figure 2b. Imidacloprid, trans-configured neonicotinoid (Schindler, 2021)

Figure 2. The structural formulas of cyclozaprid and imidacloprid. To note is the cis- and trans- configurations, as circled and labelled in blue. The cis-configuration provides a distinct spatial arrangement that enhances binding affinity to insect receptors.

A few studies have researched cyclozaprid distribution in plant tissues through foliar absorption* experiments, meaning some leaves were sprayed with the insecticide (X. Cheng et al., 2019; X. Cheng, Zhang, Shao, et al., 2022). These studies found that although most of the cyclozaprid remained in the treated leaves, some were transported to the non-treated leaves, roots, and stems (X. Cheng et al., 2019; X. Cheng, Zhang, Shao, et al.,

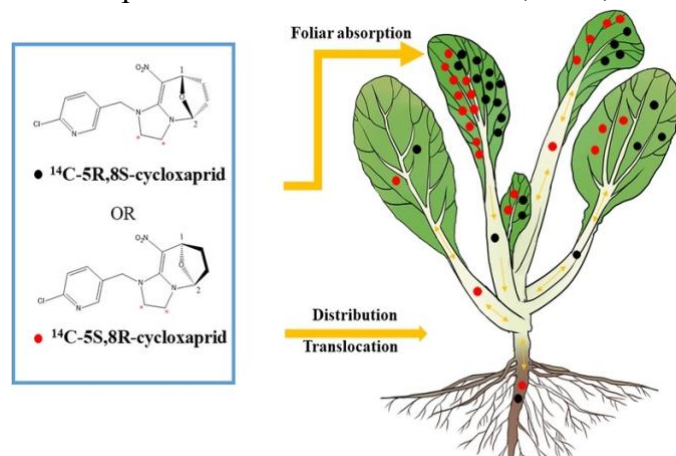


Figure 3. Distribution and translocation of enantiomers of cyclozaprid after foliar absorption in Chinese cabbage (X. Cheng et al., 2019).

2022). Other studies focused on cyclozaprid in the soil have found that after 100 days, the extractable residue of cyclozaprid decreases immensely; 70-85% of the initially applied cyclozaprid enantiomers* formed bound residues, meaning they became non-extractable (X. Cheng, Zhang, Wang, et al., 2022; Liu et al., 2015). In other words, cyclozaprid became chemically bound to soil particles in a way that could not easily be removed (Liu et al., 2015). As a result, it becomes difficult for the pesticide to be washed away with water runoffs such as rain, instead persisting in the soil, potentially affecting plants and organisms living in the soil.

Health Risks of Neonicotinoids

Neonicotinoids are known to be environmentally persistent, having the ability to exist in soil, water, and air (Gu et al., 2023; Yuanyuan et al., 2022; D. Zhang & Lu, 2022). Therefore, the possible health risks of neonicotinoids on human health have been studied extensively. Once the pesticide has entered the human body, most of it is excreted through urine; however, some may distribute throughout the body and cause adverse effects (Tao et al., 2019; D. Zhang & Lu, 2022). Various studies correlate neonicotinoids with developmental toxicity, reproductive toxicity, neurotoxicity, hepatotoxicity*, genotoxicity*, and more (Y. Chen et al., 2023; X. Li et al., 2022; Thompson et al., 2020; Tsegay et al., 2024). Neonicotinoid insecticides have also been found to cause cancer through multiple mechanisms (X. Li et al., 2022; Thompson et al., 2020; Y. Wang et al., 2022). For example, a study found that neonicotinoids stimulated a receptor called the G protein-coupled estrogen receptor (GPER), which is heavily involved in breast cancer development. As a result, the insecticides increased the migration and proliferation (rapid growth) of 4T1 breast cancer cells (X. Li et al., 2022).

While neonicotinoid insecticides aim to bind to nicotinic acetylcholine receptors* in insects, they can also bind to acetylcholine receptors in human mitochondria (S. Li et al., 2021; Skok, 2022; Xu et al., 2022). These receptors are found in the outer mitochondria membrane, and they regulate the release of reactive oxygen species* (ROS) (Skok, 2022). When the acetylcholine receptors are activated due to neonicotinoid exposure, ROS are produced in excess, which decreases cytosolic pH, causing an excessive intake of Ca^{2+} into the cytoplasm (S. Li et al., 2021; Martelli et al., 2020). As a result, cellular functions are disrupted, and it can lead to oxidative stress or even cell death (X. M. Cheng, Hu, Yang, et al., 2022; Hajam et al., 2022; S. Li et al., 2021). Oxidative stress refers to elevated levels of ROS in cells, which can damage lipids, proteins, and DNA (X. M. Cheng et al., 2022). In particular, it can cause mutations in DNA by altering DNA bases or breaking the double helix (Juan et al., 2021; Q. Wang et al., 2023). These damages can lead to DNA fragmentation, a process where DNA breaks down into smaller pieces (Juan et al., 2021; Vaughan et al., 2020). Without repair, fragmentation promote mutations that can cause uncontrolled cell growth and thus cancer (Peluso et al., 2020; Srinivas et al., 2018).

Many experiments have found correlation between traditional neonicotinoids and cancer, but with the increasing transition toward cycloxaprid usage in agriculture, the effects of cycloxaprid on the human body should be thoroughly understood as well. In particular, there are a few studies that suggest cycloxaprid triggers oxidative stress in crustaceans (Luo et al., 2024; Qi et al., 2018). However, whether cycloxaprid induces oxidative stress in humans has not been researched. More generally, whether cycloxaprid has carcinogenic effects in humans remain unknown. This gap in knowledge was confirmed

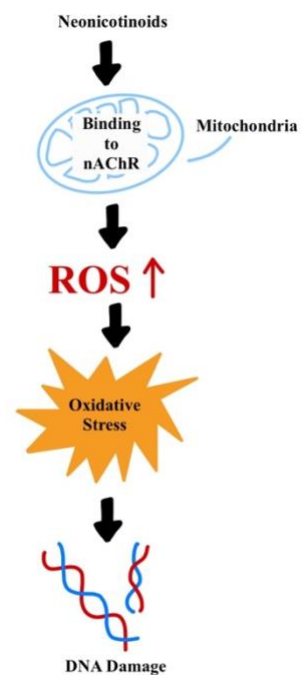


Figure 4. Process of neonicotinoids binding to nicotinic acetylcholine receptors (nAChRs) in human mitochondria and causing cancer through DNA damage.

with a thorough review of current literature using search terms such as “cycloxaprid oxidative stress,” “cycloxaprid cancer,” and “cycloxaprid carcinogen” (February 2025).

Hypothesis and Experimental Approach

Neonicotinoids, including cycloxaprid, share similar chemical structures, containing a positively charged nitro group that binds to negatively charged nicotinic acetylcholine receptors in insects (Alhalhooly & Sine, 2024; Niessen, 2021; Schindler, 2021; Q. Wang et al., 2023; Y. Zhang et al., 2023). Traditional neonicotinoids have been found to additionally bind to acetylcholine receptors in humans, triggering the release of excess reactive oxygen species (ROS) which results in oxidative stress (S. Li et al., 2021; Skok, 2022; Xu et al., 2022). This oxidative stress can lead to cellular damage, including DNA mutations and fragmentation, which, if left unrepaired, can contribute to cancer development (Juan et al., 2021; Peluso et al., 2020; Srinivas et al., 2018; Q. Wang et al., 2023). If mammalian tissue culture is exposed to cycloxaprid, there will be DNA damage in the tissue to indicate carcinogenic effects because conventional neonicotinoids induce DNA damage and cancer by binding to acetylcholine receptors, and cycloxaprid has a similar structure that binds to the same receptors in insects.

This study will research whether cycloxaprid is a possible carcinogen by examining DNA damage in mammalian tissue culture. 4T1 epithelial mouse mammary tissue culture will be exposed to a solution of cycloxaprid, and cell viability growth will be measured by calculating the number of live and dead cells per mL to observe whether DNA damage has occurred (Kemp, 2025, personal communication). A DNA laddering assay through gel electrophoresis will also be conducted to measure DNA fragmentation, an indicator of oxidative stress and cancer (Juan et al., 2021; Peluso et al., 2020; Srinivas et al., 2018; Q. Wang et al., 2023).

Methods

Organism Maintenance and Materials

This study observed the potential carcinogenic effects of the neonicotinoid insecticide cycloxaprid on 4T1 epithelial mouse mammary cells. Cells were initially preserved in a -80°C freezer. Before experimental use, cells were thawed and cultured following the BIO600 standard operating protocols. 4T1 cells were cultured under standard conditions in Dulbecco's Modified Eagle Medium (DMEM), supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin antibiotics. Cells were incubated at 37°C with 5% CO₂ in 12-well plates, each well holding a volume of 1mL. Medium was generally replaced every two days; however, when culturing a new set of cells, the medium was changed the following day to remove dimethyl sulfoxide (DMSO), which can promote cellular differentiation and alter morphology and metabolism (Y. Ding et al., 2024).

Materials used in cell culturing and experimental procedures included Corning's DMEM, Gibco's Pen/Strep, FBS, phosphate-buffered saline (PBS), and Fisher BioReagents' Dimethyl Sulfoxide. For temporary preservation of cells, such as between days when counting, 10% of DMSO was added to the growth medium,

which were placed in a cryo-box and refrozen in -80°C to protect cells from damage (Y. Ding et al., 2024; Yao & Matosevic, 2021). DMSO also served as the solvent to prepare stock solutions of cyclozaprid, which was obtained from A Chemtek and kept in a -20°C freezer in powder form. For counting purposes, cells were detached using Gibco's 0.05% Trypsin-EDTA solution and dyed with Gibco's Trypan Blue Solution to visualize dead cells. DNA extraction for the DNA laddering assay was conducted by using the Promega Wizard® Genomic DNA Purification Kit.

Determining Dosage Behavior

To assess cyclozaprid's effects on cell growth and DNA damage, two experiments were conducted using varying concentrations. The two experiments followed the same experimental processes but tested differing ranges of cyclozaprid concentrations. The first was a range-finding experiment with cyclozaprid concentrations between 0µM (control) and 1.36µM. Based on these initial results, a subsequent experiment employed a narrower range of concentrations.

Determining and Preparing Cyclozaprid Concentrations

The first experiment included a 0µM control group. Cyclozaprid concentrations were selected based on realistic environmental exposure from treated cabbage leaves, which peaked at around 1850nmol/g of cabbage (X. Cheng et al., 2019). Using the USDA serving size recommendation 70 grams of cabbage per serving, an environmental exposure of roughly 130 µmol of cyclozaprid per serving was calculated (USDA, n.d.). This is shown in the following calculation:

$$\frac{1850 \text{ nmol}}{1 \text{ gram of cabbage}} \times \frac{70 \text{ grams of cabbage}}{1 \text{ serving}} = 129,500 \frac{\text{nmol CYC}}{1 \text{ serving}} = 129.5 \frac{\mu\text{mol CYC}}{1 \text{ serving}} \approx 130 \frac{\mu\text{mol CYC}}{1 \text{ serving}}$$

Given that the plates for the cells had wells each holding 1mL, the initial goal was to prepare a stock solution 1000x the desired concentration of 130µmol/1mL. However, solubility concerns influenced the preparation methods, as the solubility of cyclozaprid in water was only 1616.19mg/L (S. Zhang & Jiang, 2023). Cyclozaprid was subsequently dissolved in DMSO, which exhibited higher solubility compared to water (Cui et al., 2018b). Given these constraints, a stock solution of 136mM was prepared by dissolving 2.2 mg of cyclozaprid powder in 50 µL of DMSO. The vial was mixed vigorously using a vortex until the powder fully dissolved in the liquid and no residues could be seen.

Exposing Cells to Cyclozaprid

For cell culture experiments, the cells were grown in 12 well plate, each well holding a volume of 1mL. Each well was filled with 50,000 cells, which was 23.5 µL of cells, though this volume may vary for each experiment. Then, to total up the volume to 1mL, 976.5 µL of growth medium was added. Cells were subsequently exposed to three cyclozaprid concentrations, each with 4 biological replicates: a control group with 0µM of cyclozaprid,

a low concentration containing 1 μL of the cyclozaprid stock resulting in 0.136 μM , and a higher concentration with 10 μL of the stock solution resulting in 1.36 μM . Each of these amounts of cyclozaprid stock solution (0, 1 μL , 10 μL) was pipetted directly into the medium. The cells were then incubated for 48 hours.

After the two days of incubation, the growth medium in each of the biological replicates was pipetted out into 1.5mL vials. This ensured that any dead cells floating in the growth medium, unable to attach to the floor of the well, would be preserved and later counted. Then, the remaining live cells, adhered to the walls, were detached via trypsinization following the BIO600 standard operation protocol. After trypsin treatment and incubation for 3 minutes, the plate was tapped a few times and checked under the microscope to ensure detachment. Most cells were not floating yet, so the cells were incubated again for another three minutes.

When centrifuging the cells to pellet them, each of the vials had a balance on the opposite side. After centrifuging, the tubes were handled extremely gently to prevent disturbing the pellets. When pipetting the medium/trypsin mixture, the vial was also tilted toward the opposite side of the pellet. This ensured that the pellets were not disturbed, and the cells were not accidentally pipetted out. After the medium was resuspended in 1mL of medium, they were also put into 1.5mL vials labelled L for live cells.

Both live and dead cells were counted following the BIO600 standard operation protocols using a hemocytometer under a microscope. For accuracy, each pipette vial was counted in two technical replicates, each replicate involving three squares in the hemocytometer. To count the cells, 20 μL of the cells and 20 μL of Trypan Blue dye was pipetted to a smaller vial. Before pipetting out the 20 μL of cells, however, the original vials were resuspended by using a larger pipette to pipette up and down. This prevented the risks of extracting erroneous cell counts affected by the pull of gravity. When adding cells to the chamber of the hemocytometer, the chamber was placed on the hood tabletop, and the cover was gently held down, ensuring no spilling of the cells occurred. Finally, under the microscope, only two out of four sides of each of the squares were counted to maintain consistency and minimize errors in the results (Kemp, 2025, personal communication).

Follow up experiment

Due to the observed substantial drop in live cell counts between 0 μM and 0.136 μM , a second experiment further explored concentrations between 0 μM and 0.136 μM . The same methods as the previous range-finding experiment were followed, except for the concentrations of cyclozaprid. For this experiment, the stock solution (0.0136mM/mL) was prepared using 2.2mg of cyclozaprid dissolved in 500 μL of DMSO. Four concentrations were tested. First, for the control group, 0 μM of CYC was added, but 10 μL of DMSO was added to distinguish solvent effects. The other three concentrations used 1 μL , 5 μL , and 10 μL of the stock solution to achieve concentrations of 0.0136 μM , 0.068 μM , and 0.136 μM , respectively.

DNA Laddering Assay

To identify DNA fragmentation indicative of oxidative stress and apoptosis, a DNA laddering assay was conducted using agarose gel electrophoresis (Juan et al., 2021; Peluso et al., 2020; Saadat et al., 2015; Srinivas et al., 2018). Fragmented DNA was isolated and separated based on size through electrophoresis in a 1.5% agarose gel. Following BIO600 standard operating protocols, 4T1 epithelial mouse mammary cells were cultured, exposed with varying levels of cycloaprid ($0\mu\text{M}$, $0.0136\mu\text{M}$, $0.068\mu\text{M}$, and $0.136\mu\text{M}$) and trypsinized. The DNA from these cells were then extracted using the Promega Wizard® Genomic DNA Purification Kit. The extraction was conducted following the Technical Manual for the Wizard® Genomic DNA Purification Kit's DNA extraction protocol, which was chapter 3D: Isolating Genomic DNA from Tissue Culture Cells and Animal Tissue (Promega Corporation, 2023). Of chapter 3D, steps 1 and 5-14 were followed for 4T1 epithelial mouse mammary cells. Whenever the protocol said to vortex the tubes, I flicked the tubes with my index finger instead, to prevent damaging DNA. Furthermore, on step 8, the solution was inverted back and forth gently for ten minutes to ensure that the strands of DNA would form a very visible mass.

For the gel electrophoresis, a 1% agarose gel was created with GelRed following BIO600 standard protocols. Over the gel, Tris-Acetate-EDTA (TAE) buffer was poured over. As the gel cooled in the fridge, the different DNA samples were created. Concentrations of DNA was measured using the Thermo Scientific Nanodrop Lite spectrophotometer, and for each test group, 100ng of DNA was calculated and put into vials. Then, water was added to the vial to bring the total volume of the vial to be $10\mu\text{L}$. Finally, to each vial, $2\mu\text{L}$ of loading dye was added, making the total volume of the vial $12\mu\text{L}$.

The gel electrophoresis was conducted following Saadat et al.'s protocol, but I substituted SYBR-Safe DNA gel stain with GelRed, a sensitive and safer alternative (Saadat et al., 2015; Zheng et al., 2025). In the first well, $5\mu\text{L}$ of the DNA ladder was added.

Then, each subsequent ladder was filled with DNA in ascending order of cycloaprid concentrations to allow for easier comparison between ladders. Before loading each well with the $12\mu\text{L}$ of solution that included around 100 ng of DNA, water, and loading dye, the solution was pipetted up and down ten times in the vial to ensure that the dye was fully mixed in to the DNA. Then, for each well, the pipet tip was held over the well, with my non-dominant hand stabilizing the pipette.

Gently lower the pipette into the running buffer, putting the pipette tip into the well; this should be deep but not too deep that it protrudes through the bottom of the well. One way to check this is by wiggling the pipette a little

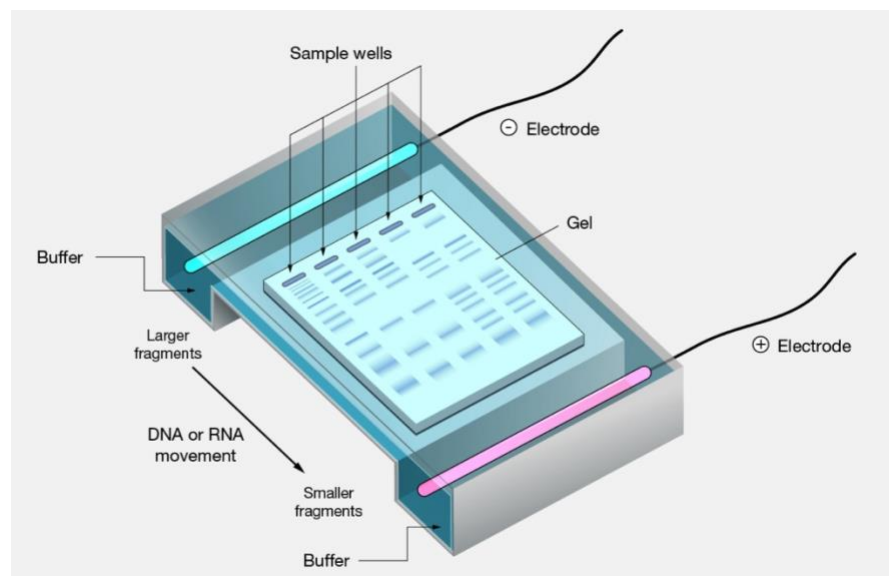


Figure 5. An image of a gel electrophoresis system, with each component labelled. The electric field pulls the negatively charged DNA from the anode to the cathode, with smaller fragments travelling further distances (Smith, 2025).

to see if the gel moves. After pipetting all the DNA into respective wells, slide the cover over, ensuring that the top of the well, closer to where the DNA starts, is the black end (blue in Fig. 5), which represents the anode. The red part (pink in Fig. 5), the cathode, should be at the opposing end, because it should attract DNA, which is negatively charged (Wong & Liu, 2021). The gel was then plugged into a power supply and ran at 100mA and ran for thirty minutes. To ensure that the gel is running correctly, check for bubbles at the edges of the tank near the electrodes. After the gel has completed running, unplug the device, and gently remove the gel from the tank. The gel was visualized under a UV bulb using the Analytik Jena UVP UV/White Transilluminator.

Mathematical Analysis/Statistics

To determine live and dead cell counts (cells/mL) for each cyclosporin concentration, three to four biological replicates were used. Each biological replicate had two technical replicates. For each technical replicate, three squares of a hemocytometer were counted and averaged. The averages from the two technical replicates were combined to get one value per biological replicate. Finally, results from all biological replicates were averaged to obtain the overall cell count for each cyclosporin concentration.

To convert this average count into a cell concentration (cells/mL), the number was multiplied by a dilution factor of 2, accounting for the addition of the equal volume of Trypan Blue and then multiplied by a factor of 10^4 to convert the units from 0.1μL to 1mL. This calculation is shown below:

$$\text{cells concentration} \left(\frac{\text{cells}}{\text{mL}} \right) = \text{cells per CYC concentration} \times \text{dilution factor} (2) \times 10^4$$

This calculation was completed for both live and dead cells for each of the cyclosporin concentrations and then graphed to examine a trend.

For the DNA laddering assay, the resulting ladders were analyzed using ImageJ software. The pixel intensities were compared to look at the relative amounts DNA fragmentation between the DNA of cells exposed to 0.068μM CYC and 0.136μM CYC. To compute the change in amounts of DNA fragmentation, pixel intensities for the two ladders were found as well as the pixel intensity of the well background. Then, the background intensity was subtracted, and the percent change between the two test groups was calculated as shown in the following calculation: $\frac{\text{Intensity}_{0.136} - \text{Intensity}_{0.068}}{\text{Intensity}_{0.068}} \times 100\%$.

Results

To determine whether cycloxabrid is a carcinogenic agent, 4T1 epithelial mouse mammary cells were exposed varying levels of cycloxabrid (0 μ M, 0.0136 μ M, 0.068 μ M, 0.136 μ M, and 1.36 μ M) for 50 hours. The results exhibited a clear and dependent trend for both live and dead cell counts (Fig 6). With the dead cells, the count increased in an almost linear fashion as cycloxabrid concentrations increased, while the live cell count showed a rapid, exponential decline.

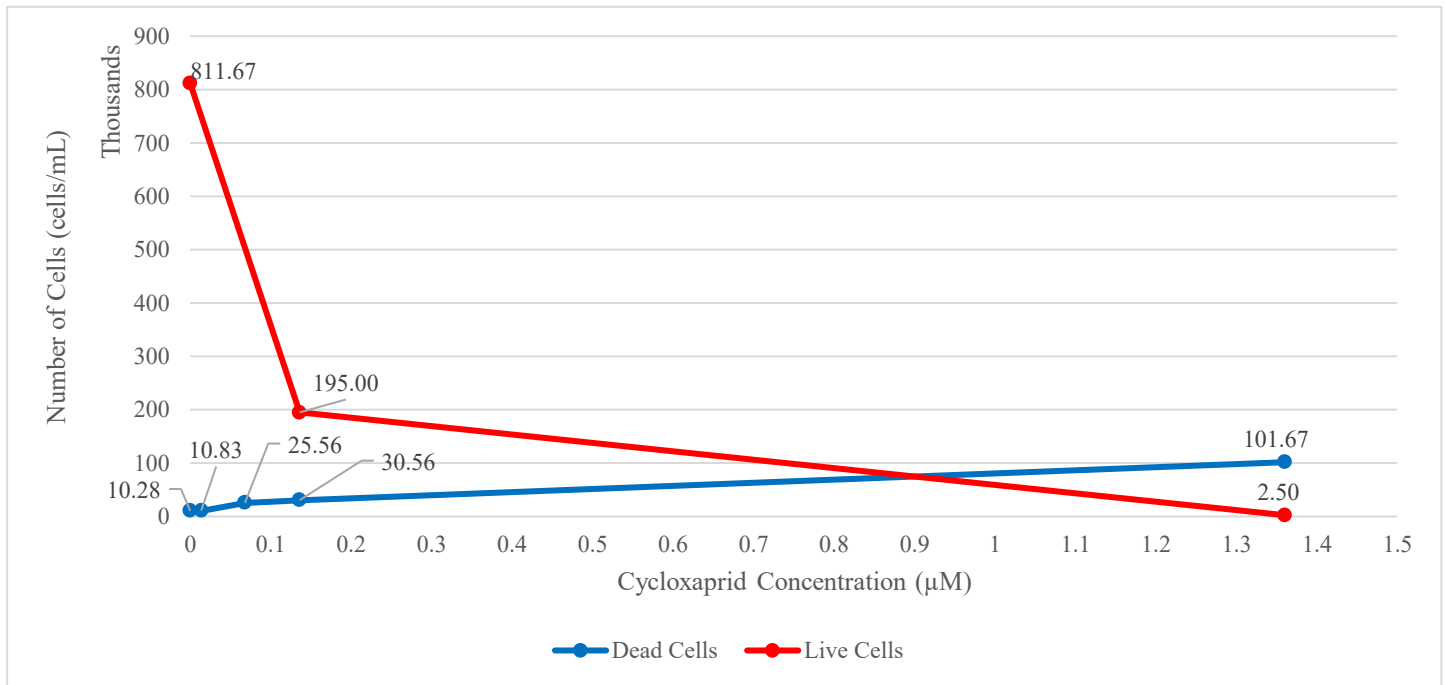


Figure 6. In order to see whether cycloxabrid was a carcinogenic agent, 4T1 epithelial mouse mammary cells were exposed varying levels of cycloxabrid (0 μ M, 0.0136 μ M, 0.068 μ M, 0.136 μ M, and 1.36 μ M) for 50 hours. The average (6 technical replicates for 3-4 biological replicates) concentration of cells (cells/mL) were recorded. Each well (biological replicate) was initially seeded with 50,000 cells. In the graph above, the blue line represents the average number of dead cells per milliliter, while the red line represents the average number of live cells per milliliter. The data clearly indicates that increased cycloxabrid levels are toxic toward cells, as shown by the near-linear increase of dead cell counts. This suggests that cycloxabrid exposure induces molecular mechanisms which trigger cellular death. On the other hand, the decrease in live cell counts reflect a lower doubling rate which suggests inhibition of the cell cycle. The doubling rate for 4T1 cells under standard conditions ranges from 12.1-15.1 hours, as reflected in the 0 μ M concentration group which has a doubling rate of 12.5 hours. However, at lower concentrations, the live cell count decreased sharply, reflected by a doubling rate of around 20 hours for the 0.136 μ M group, and a live cell count that was 5% of the original cell count for the 1.36 μ M concentration.

In the untreated control group (0 μ M), the average dead cell count was relatively low at 10,280 cells/mL, with live cells significantly outnumbering them at approximately 811,670 cells/mL. Under standard conditions, the doubling rate ranges from 12.1-15.1 hours, which was reflected in this control group as the live cells in this concentration a doubling rate of 12.5 hours. However, at the next concentration of 0.136 μ M, the live cell count decreased sharply with 195,000 cells/mL, reflecting a slower doubling rate of around 20 hours for the 0.136 μ M group. At the 1.36 μ M concentration, the live cell count continued to decrease, with a live cell count of 101,670 cells/mL, which only makes up 5% of the original cell count for the control group.

On the other hand, the dead cell count continued to increase, with a 5.35% increase between the 0 μ M and 0.0136 μ M concentration,

At 0.136 μ M, the dead cell count surged roughly threefold compared to the original dead cell count at 0 μ M. Finally, at the highest concentration of cycloxabrid tested (1.36 μ M), the number of dead cells increased to

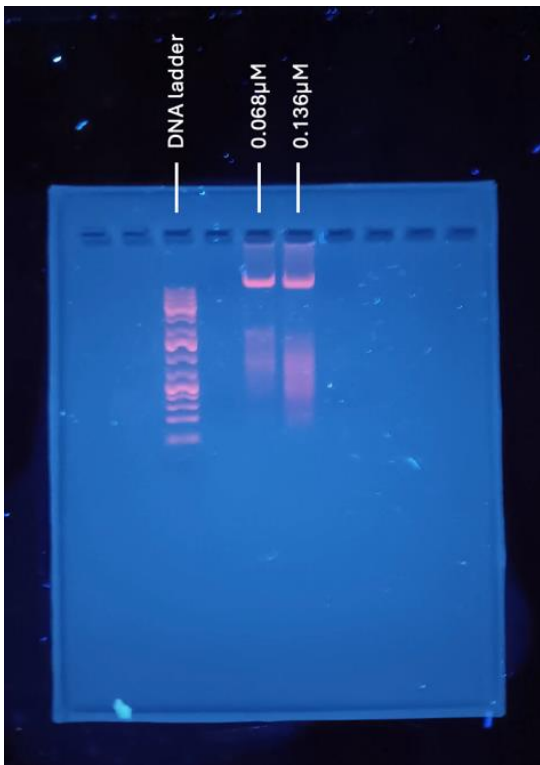


Figure 7. To determine whether 4T1 epithelial mouse cells treated with cyclohexaprid were dying through apoptosis or necrosis, a DNA laddering assay through gel electrophoresis was conducted. The DNA was isolated using the Wizard Genomic DNA Purification Kit protocol and quantified with a spectrophotometer to load approximately 100ng of DNA per well, alongside a DNA ladder. A 1% agarose gel with GelRed was created and the electrophoresis was run with TAE buffer at 100mA for thirty minutes. Degraded DNA was then visualized under UV light. Pixel intensity analysis showed that DNA degradation increased by around 28.576% between the 0.068µM and 0.136µM cyclohexaprid concentrations. In both wells, it is mostly smearing but also some laddering; this means that both apoptosis and necrosis are likely occurring. These results are consistent with the data from figure 1, showing that as cyclohexaprid concentrations increase, so does DNA damage and consequent cell death.

that with heightened cyclohexaprid exposure, cells would grow uncontrollably as a result of cancer. A follow-up experiment tested a narrower range of cyclohexaprid concentrations and found a similar trend in the number of dead cells. Moreover, the second experiment's data showed an increase in quantity of DNA fragmentation as concentrations of cyclohexaprid increased, suggesting that increased cyclohexaprid concentrations would induce DNA fragmentation (Fig 7). As DNA fragmentation is a key characteristic of apoptosis, it would show that the cells were not replicating uncontrollably, but rather dying likely due to apoptotic mechanisms (Basnakian & Moore, 2021).

Inhibition of Cell Cycle and Growth

The first experiment examined the viability of cells following exposure to varying levels of cyclohexaprid (0µM, 0.0136µM, 0.068µM, 0.136µM, and 1.36µM). Results showed that as cyclohexaprid concentrations

a whopping 101,670 cells/mL, heavily outnumbering the live cell count which lay at a low 2,500 cells/mL. Overall, cyclohexaprid exposure reduced 4T1 cell viability, demonstrating an inverse relationship between cyclohexaprid concentration and live cell count, along with a corresponding increase in dead cell count.

Subtracting the pixel intensity of the well background, the 0.068µM DNA had 473,536 pixel intensity, while the 0.136µM DNA had 608,854.4 pixel intensity. Thus, the percentage increase of DNA fragmentation from the 0.068µM DNA to the 0.136µM DNA was $\frac{608,854.4 - 473,536}{473,536} \times 100\% \approx 28.576\%$.

Discussion

Conclusion Regarding Hypothesis

In examining the effects of the neonicotinoid cyclohexaprid on 4T1 epithelial mouse mammary cells, I hypothesized that cyclohexaprid exposure would induce DNA damage, leading to cancer and uncontrollable cell growth. However, this hypothesis was refuted by two experiments. The first experiment examined the amount of cell growth (live and dead cell counts) after exposure to cyclohexaprid. The results showed that the number of live cells drastically decreased with exposure to increased levels of cyclohexaprid, while the number of dead cells increased proportionally with the amount of cyclohexaprid (Fig 6).

This means that cell viability decreased, contradicting the hypothesis

increased, cell viability decreased drastically (Fig 6). Under normal conditions, 4T1 cells double approximately every 12.1 to 15.1 hours (Simões et al., 2015). In the absence of cycloxabrid (control group, 0 μ M), the cells increased from an initial concentration of 50,000 to 811,670 cells/mL, after 50 hours, reflecting a doubling time of roughly 12.5 hours, consistent with the expected growth rate. This confirms that the cells were healthy and growing normally in the control condition.

However, the treatment groups exposed to higher levels of cycloxabrid showed drastic reductions in live cell concentrations. At the 0.0136 μ M concentration, there were 195,000 cells/mL which reflected a doubling time of around 20 hours (Fig 6). At the 0.136 μ M concentration, the live cell count was a mere 2,500 cells/mL, which is 5% of the initial number of cells that were seeded, meaning most cells died before even getting through the cell cycle and mitosis (Fig 6). The evidently slower doubling rate of the cells suggest that cycloxabrid exposure inhibited the cell cycle.

Cycloxabrid's interaction with mitochondrial nicotinic acetylcholine receptors (nAChRs) likely explains the reasons for the decline in live cell counts (S. Li et al., 2021; Skok, 2022; Xu et al., 2022). Cycloxabrid has a positively charged nitro group which binds to the negatively charged nAChRs located in the outer membrane of the mitochondria (Alhalhooly & Sine, 2024; Niessen, 2021; Schindler, 2021; Skok, 2022; Q. Wang et al., 2023; Y. Zhang et al., 2023). Activation of these nAChRs by cycloxabrid can disrupt the regulation of the release of reactive oxygen species (ROS), causing elevated levels of ROS in cells; this disrupts cellular functions and damages lipids, proteins, and DNA (X. M. Cheng, Hu, Yang, et al., 2022; Hajam et al., 2022; S. Li et al., 2021).

Furthermore, oxidative stress and elevated ROS production induced by cycloxabrid significantly affect the cell cycle at a molecular level. Under normal conditions, the cell cycle progresses through four stages: G₀/G₁ (gap 1), S (DNA synthesis), G₂ (gap 2), and M (mitosis), with multiple checkpoints that ensure proper cell division as shown in Figure 8 (L. Ding et al., 2020). This progression is primarily regulated by cyclins and Cyclin Dependent Kinases (CDKs) (L. Ding et al., 2020; Łukasik et al., 2021). When oxidative stress occurs, excessive ROS directly damages DNA, triggering the

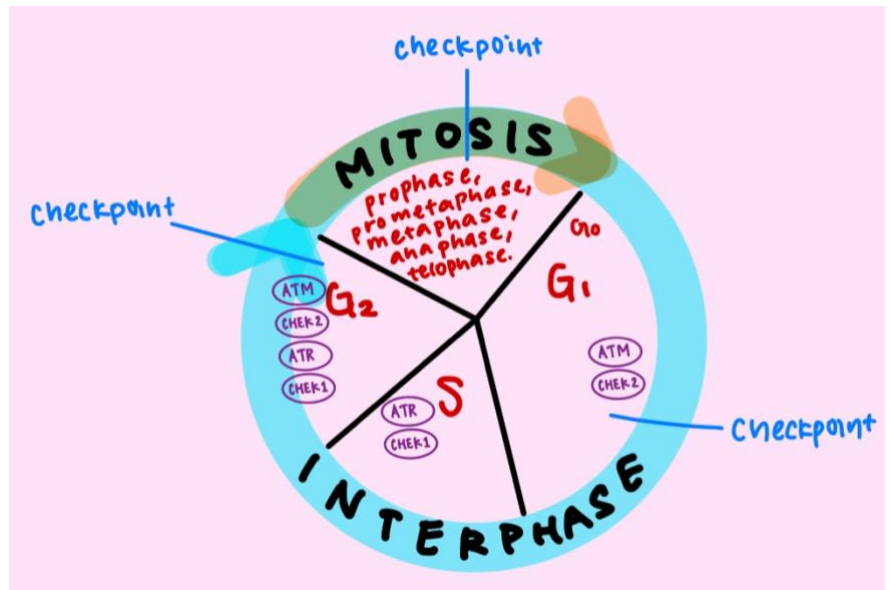


Figure 8. A depiction of the cell cycle with the checkpoint kinases labelled. ATR phosphorylates CHEK1, and ATM phosphorylates CHEK2. These kinases regulate cell cycle arrest and progression through the cycle.

DNA damage response (DDR) (Clay & Fox, 2021; Juan et al., 2021; Q. Wang et al., 2023). Activation of the DDR network engages many essential regulators such as the ATR (Ataxia telangiectasia Rad3-related) and ATM (Ataxia telangiectasia mutated), which phosphorylate Checkpoint kinases CHEK1 and CHEK2, respectively

(Clay & Fox, 2021; Gentric et al., 2021; R. X. Huang & Zhou, 2020). Activated CHEK 1 inhibits CDKs, leading to a halt in cell cycle progression because CDKs are vital in driving the cell through checkpoints and the cell cycle (Blackford & Stucki, 2020; Clay & Fox, 2021). On the other hand, ATM and CHEK 2 both regulate the tumor suppressor gene p53, which can also regulate cell cycle arrest by halting the cell cycle at checkpoints to allow for DNA repair (Clay & Fox, 2021).

Cell Death

The first experiment also found that the number of dead cells rose in a nearly linear relationship with increased cyclophosphamide concentration (Fig 6). This result contradicts the initial hypothesis, which predicted carcinogenic effects characterized by uncontrolled cell growth (Maadi et al., 2022; Mercadante & Kasi, 2023). Instead, cells underwent significant cell death, suggesting that cyclophosphamide was highly cytotoxic rather than carcinogenic. With cancer, proliferation occurs with damaged cells that avoid apoptosis, allowing mutations to form, leading to cancer development (Maadi et al., 2022; Mercadante & Kasi, 2023). The clear cell death observed here suggests that cells failed to bypass the checkpoints, instead proceeding to cell death.

A possible explanation for the cytotoxic effect lies in cyclophosphamide's interaction with mitochondrial nicotinic acetylcholine receptors (nAChRs), just like with the live cells (S. Li et al., 2021; Skok, 2022; Xu et al., 2022). Through the activation of these nAChRs and the excess of ROS, there is an imbalance between ROS production and defense mechanisms; this is known as oxidative stress, and it is recognized as a trigger for apoptosis or necrosis (X. M. Cheng, Hu, Yang, et al., 2022; Hajam et al., 2022; S. Li et al., 2021; Zhuang et al., 2020). For example, oxidative stress may damage cellular components and DNA, causing mutations (Juan et al., 2021; Wang et al., 2023). If these mutations do not get fixed, similar to the mechanism for live cells, the cell may undergo cell cycle arrest and result in apoptosis if the DNA is not able to be repaired. Thus, cyclophosphamide's influence in elevating ROS levels explains the observed increase in cell death after exposure to the chemical.

DNA Laddering Assay and DNA Fragmentation

To understand the mechanisms for the observed cell death, DNA fragmentation was examined via a DNA laddering assay, which is a classic characteristic of apoptosis (Costigan et al., 2023; Terabayashi et al., 2020). This is because when apoptosis occurs, caspase-activated DNase cleaves DNA in an orderly pattern to create DNA fragments, which can be visualized as a ladder in gel electrophoresis (Y. Chen et al., 2024; Ha & Park, 2022). On the other hand, necrotic cell death is typically not associated with caspase activation; rather, the cell membrane breaks down (Khalid & Azimpouran, 2023). This causes the cell to swell and rupture, with intracellular contents spilling out of the cell into surrounding tissues, as shown in Figure 9 (Khalid & Azimpouran, 2023).

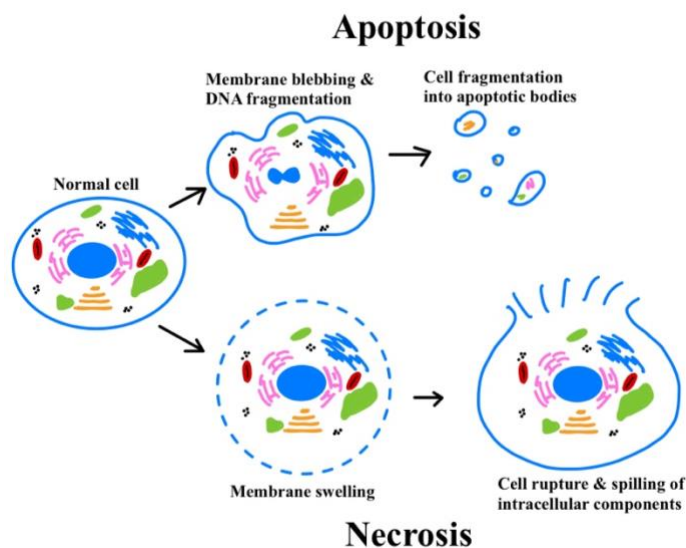


Figure 9. The difference between apoptosis and necrosis. During apoptosis, DNA is cleaved in an orderly manner, while with necrosis, the cell membrane ruptures, resulting in a spill of intracellular components.

mechanisms of cell death resulting from exposure to cycloaprid.

Molecularly, elevated ROS as a result of cycloaprid may damage DNA, causing mutations in DNA by altering bases or by breaking the double helix (Juan et al., 2021; Q. Wang et al., 2023). These damages may trigger a wide range of responses, including the activator of the tumor suppressor protein p53 and the upregulation of caspases which leads to DNA fragmentation (T. Shi et al., 2021). As apoptotic signaling pathways are amplified, as ATM, ATR, and the checkpoint kinases CHK2 and CHK1 makes the cell enter apoptosis (Juan et al., 2021; Vaughan et al., 2020; Waterman et al., 2020).

On the other hand, increased ROS production due to cycloaprid can also damage cellular components such as the mitochondria, leading to mitochondrial dysfunction (Jurcău et al., 2022; S. Shi et al., 2021). As the mitochondria is the main site for ATP production in cells, dysfunction leads to a high depletion of ATP (F. Chen et al., 2024; Hyatt & Powers, 2021; S. Shi et al., 2021). Lack of ATP leads to failure of ATP dependent pumps for sodium and calcium ions; as a result, the cells become imbalanced with ions, and the cell eventually loses its membrane integrity, ultimately rupturing the cell (Dunn & Grider, 2023; S. Huang et al., 2024). As the membrane ruptures, the intracellular contents spill out of the cell into surrounding tissues, and necrosis occurs (Khalid & Azimpouran, 2023; Mangalmurti & Lukens, 2022). Therefore, the experimental findings indicate that cycloaprid induces oxidative stress, which significantly triggers both apoptosis and necrosis.

Sources of Error/Limitations

Due to the small sample size of the two data sets used in this study, there are limitations to the accuracy of the exact numbers. Furthermore, there are limitations with the amount of cycloaprid concentrations tested. In the follow-up experiment that examined cell viability with a narrower range of concentrations (0μM, 0.0136μM, 0.068μM, 0.136μM), there was an error when trypsinizing and harvesting the cells. After centrifuging the 1.5mL

Furthermore, in necrosis, DNA is not cut in an orderly manner, so instead of DNA fragmentation and a consequent clear ladder, the DNA is smeared in electrophoresis (Didenko et al., 2003; To & Shu, 2022).

In the experimental results, increasing cycloaprid concentrations from 0.068μM to 0.136μM led to a measurable increase in DNA degradation by approximately 28.576% (Fig 7). The data suggests that higher cycloaprid levels lead to heightened DNA damage. Furthermore, examining the patterns of the DNA laddering assay highlights that there is clearly DNA smearing as well as some laddering (Fig 7). This suggest that both necrosis and apoptosis occurred as the

vials to pellet the cells, a 1mL pipette was used to remove the residual liquid. However, using a 1mL pipette was too large of a pipette, and the cell pellets were disturbed, resulting in most of the cells being pipetted out along with the excess liquid. Therefore, when counting the number of live cells, the counts were very low. For example, for the control group with 0 μ M of cycloxaprid, there was an average of 1,000 cells, though there should have been 800,000 cells according to the doubling time of 4T1 cells. Thus, the data for the live cell counts in a narrower range of cycloxaprid concentrations was omitted.

With the DNA laddering assay, the data was limited by its sample size especially as the DNA for two concentrations (0 μ M and 1.36 μ M) were lost and not able to be tested. For the DNA for the 1.36 μ M concentration, it was pipetted out during the DNA extraction process. Additionally, the DNA for the control group (0 μ M) was misplaced and lost, so I was not able to compare the resulting ladders with a control sample. The remaining four concentrations (0.0136 μ M, 0.068 μ M, 0.136 μ M, and 0.68 μ M) were tested, but during the DNA laddering assay, no DNA could be visualized for the 0.0136 μ M and 0.68 μ M concentration. No plausible explanation could be found.

Broader Impacts

With the current problems surrounding climate change, especially with the rise of pest populations and pesticide resistance, cycloxaprid is a novel and promising insecticide, predicted to be the next most prevalent neonicotinoid insecticide (Cui et al., 2020; Dong et al., 2022; Duchenne-Moutien & Neetoo, 2021; Luo et al., 2024). Because pesticides are applied in many ways, such as through spraying plant leaves and into the soil, cycloxaprid can be found in a range of environments including aquatic ecosystems, soils, and human consumed plants (X. Cheng et al., 2019; Kaur et al., 2024; Tao et al., 2019). Furthermore, cycloxaprid is highly persistent in the environment; for example, when treating leaves with cycloxaprid, the transported to the other leaves, roots, and stems (X. Cheng et al., 2019; X. Cheng, Zhang, Shao, et al., 2022). Furthermore, cycloxaprid forms bound residues in soils, becoming difficult to be washed away with water runoff (X. Cheng, Zhang, Wang, et al., 2022; Liu et al., 2015). After making its way into human bodies, cycloxaprid also has high potential to bioaccumulate in animal tissue (Li, 2020). With the prevalence of cycloxaprid in the environment and potentially in human bodies, it is essential to fully understand the extent to which it may affect the human body.

This study suggests that cycloxaprid may not induce cancer, but it can cause a decrease in cell viability through oxidative stress, DNA fragmentation, and consequently, cellular apoptosis. Cell death resulting from oxidative stress and DNA damage poses many serious health risks for humans, leading to tissue dysfunction and various chronic diseases. Some examples include neurodegenerative diseases like Alzheimer's and Parkinson's disease, cardiovascular and kidney diseases, and diabetes mellitus, and more (Leyane et al., 2022; Newton et al., 2024; Park et al., 2023). As these diseases affect tens of millions of people globally, understanding the mechanisms by which environmental contaminants may induce cellular toxicity is crucial for assessing their health risks and preventing adverse health outcomes (Ortega et al., 2023).

Next Steps

This study suggests that cycloxaprid can inhibit cell cycle progression and promote cellular death through oxidative stress, rather than causing uncontrollable cell growth and cancer in 4T1 epithelial mouse mammary cells. Future research should continue to examine various aspects of cycloxaprid's effects on cells to better understand its biological impacts and potential health effects. As this study examines a small sample size and range of cycloxaprid concentrations, additional levels of cycloxaprid, including extreme concentrations, should be tested with much larger sample sizes to determine whether this is a reproducible finding and if there are true biological trends. If this conclusion is fully supported with more data, further experiments should test to quantify ROS amounts and look at specific cell death pathways. Furthermore, experiments should test whether cycloxaprid truly interacts with nAChRs; looking at ROS quantification or mitochondrial membrane potentials could provide a better understanding of how cycloxaprid affects cells and verify the molecular mechanisms of oxidative stress. This research should also extend to other cell lines, including non-cancerous cells to observe whether cycloxaprid exhibits similar toxicity to normal cells. Furthermore, considering cycloxaprid's environmental persistence and potential for bioaccumulation, its long-term effects should also be fully understood. Future experiments should not only quantify cycloxaprid's presence in various environments but also look at its long term effects in animal cells and tissue to understand its bioaccumulation and toxic effects following chronic exposure. Finally, research should examine preventative strategies against cycloxaprid's toxic effects. These studies could inform risk assessments and health guidelines, especially in agricultural contexts where human and animal exposure would be at its highest. As cycloxaprid gains prevalence, it is necessary for continued research around the biological impacts of cycloxaprid to create safe and informed decisions and use of the insecticide.

Disclosures/Statement of Ethics

There are no conflicts of interest to be stated.

Positionality Statement

Before presenting these findings, I would like to acknowledge my perspective as a Japanese international student studying in America. I grew up with Japanese parents and their beliefs, but I lived in Singapore, where I studied three languages. Especially due to the blend of nationalities and identities in Singapore, I've been exposed to a multicultural background with diverse academic and living environments. Now as a senior in high school in America, I've faced new ideas, different approaches to scientific research, and a wider and more global perspective.

Acknowledgements

I would like to thank my mentor Dr. Kemp for the tremendous amount of help I've received throughout this research with conferencing about experimental methods, feedback on manuscript drafts, and emotional support. I would also like to thank Mr. Blake for assistance with reviewing literature in the early stages of this study. Furthermore, I would like to thank Ms. Shahshahan for supporting me every day, especially with tissue culture work, helping me find solutions when I make mistakes, and just being overall an amazing mentor. I would also like to thank Jaceil for peer reviewing my paper and giving me a lot of good feedback. Finally, I would like to give a big shout out to my Bio600 class for the fun, encouragement, and help with various experimental methods.

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